Skills Summary

Diagnosing Causal Claims and Biased Study Designs

Use this reference while completing your worksheet or when studying.

Skill 1 — Coverage and response rate

Two different fractions, two different faults.

- **Coverage** = (units the frame can reach) ÷ (units in the population). Low coverage is **undercoverage**: some units have probability zero of ever being selected.
- **Response rate** = (units that answered) ÷ (units contacted). Low response is **nonresponse bias** when the silent units differ from the ones who answered.
- **Voluntary response** is the worst case of both: respondents select themselves, so strong opinions are over-represented by design. A larger sample fixes none of this — it shrinks the interval around a biased estimate.

Example: A survey is mailed to 300 households and 45 of them return it. Give the response rate as an exact fraction.
Step 1. The question is about who answered among those contacted, so the denominator is the number contacted, not the number in the town.
Step 2. Form 45 over 300 and reduce by the common factor 15.
The response rate is $\frac{3}{20}$, so most of the contacted households are unrepresented in the result.

Try it: A survey is mailed to 300 households and 105 return it. Give the response rate as an exact fraction.

check: $\frac{7}{20}$

Skill 2 — Comparing two groups honestly

Before comparing two group means, ask who is in each mean.

- **Attrition:** average everyone who *started*, not everyone who *finished*. Dropping the leavers biases the treatment mean upward when the leavers were doing worst.
- **Assignment:** if subjects chose their own group, the study is observational and the gap is association.
- **Confounding:** name a variable that differs between the groups and moves the outcome on its own (motivation, baseline level, income, age). **The remedy** is random assignment from a single pool, with every randomized subject kept in the arithmetic.

Example: Volunteers for a program scored 8, 2, 5; people who did not volunteer scored 4, 1, 1. Find both means and judge the claim "the program works".
Step 1. Average each group over its own members only — the comparison is meaningless if one mean quietly excludes the people who did badly.
Step 2. Add each group's three values and divide by three.
Volunteers: **5**. Non-volunteers: **2**. The gap is real, but the subjects volunteered, so it is association — motivation alone could produce it.

Try it: One group scored 10, 6, 8; the other scored 5, 4, 3. Find both means.

check: **8** and **4**

Skill 3 — Is the sample representative?

Compare the sample's composition with the population's, on a variable you already know.

- Over-representation in **percentage points** is a **subtraction**: sample share − population share.
- Dividing the two shares gives a ratio. Reporting a ratio as "points" understates a large skew and overstates a small one.
- A visible skew on one known variable is evidence that unknown variables are skewed too. Fix it by scoping the conclusion to who was actually sampled, or by drawing a probability sample from a frame that covers the target population.

Example: A mall-intercept sample is 48% students, while the town is 18% students. By how many percentage points are students over-represented?
Step 1. Percentage points compare two shares by subtraction, so take the sample share minus the population share rather than their ratio.
Step 2. Subtract: $48 - 18$.
Students are over-represented by **30** percentage points.

Try it: A sample is 62% homeowners while the town is 25% homeowners. By how many percentage points are homeowners over-represented?

check: **37**